

Statistical methods for computer science

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Sampling Distributions

Intro

Descriptive statistics

Collect, summarize, and describe the data

Inferential statistics

Draw conclusions about a **population** of interest based on **sample** data

Population

Set of all units of interest

- All voters at the next elections
- All bills issued in April

Sample

Subset of units from the population of interest

- Some voters selected for a survey
- Some bills selected for a verification

- In detail, inferential statistics is a quantitative **induction procedure** in which **uncertainty** is formally quantified (i.e., it is translated into one or more numbers)
- Uncertainty is due to **two main sources**

Sampling variability

- All samples are in principle different
- Their analysis leads to different results/conclusions
- We only observe one out of all possible samples

Measurement variability

- Repeating the measurement process on the same unit, we generally obtain different values
- Typically, we only have a single measurement per unit

Sampling

- Assume, we are interested in a given characteristic of the population, Y , and assume its probability distribution is $f(\cdot)$; that is,

$$Y \sim f(\cdot)$$

- Assume, we **randomly** draw a unit i from the population
- The value of Y for the sampled unit (Y_i) is unknown a-priori
- That is, Y_i is a **random variable** having the same distribution as that of the characteristic Y in the population

$$Y_i \sim f(\cdot)$$

- Similarly, if we draw randomly and with repetition n units from the population, we have n random variables

$$Y_1, \dots, Y_n,$$

being **independent and identically distributed** (iid) as the characteristic Y in the population

$$Y_i \stackrel{iid}{\sim} f(\cdot)$$

- After the experiment, the sample corresponds to an n -dimensional vector of realizations

$$\mathbf{y} = (y_1, \dots, y_n)'$$

An example

- 50 packs of pasta are randomly drawn from an industrial process
- The population is made by all potentially producible packs
- Characteristic of interest: $Y = \text{weight of a pack}$
- Hypothesis: $Y \sim N(\mu, \sigma^2)$
- Weight of the i -th pack: $Y_i \sim N(\mu, \sigma^2)$
- Weight of the n packs: $Y_1, \dots, Y_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$
- Once we draw a sample, we observe the weight of n different packs of pasta $y_1, \dots, y_n$

Parameters

- Typically, inference entails one or more **parameters** of the population

Formally, a **parameter** is a numerical summary describing the distribution of the characteristic Y in the population (the mean, the median, the standard deviation, etc..)

- If we assume that the distribution of Y is well described by a given parametric family of distributions (the Gaussian, the Binomial, etc..), then the parameters of such a family represents the **unknown quantities** in the population
 - If we assume $Y \sim N(\mu, \sigma^2)$, the mean and variance are the unknown parameters of interest
 - Should we know the value of μ and σ^2 , we would know everything about the distribution of Y in the population

Sampling statistics

- Each summary in the population (i.e., each parameter) has its analogous in the sample; these are known as **sample statistics**

Formally, a **sample statistic** is a numerical summary describing a sample feature

- For instance, the **population mean** of a given characteristics Y has its analogous in the sample corresponding to the **sample mean**
- The **population variance** of a given characteristics Y has its analogous in the sample corresponding to the **sample variance**
- ...

- Before observing the sample, the specific value of the characteristic of interest Y for the sampled units is unknown and so is the value of the statistic

$$Y_1, \dots, Y_n \rightarrow f(Y_1, \dots, Y_n)$$

- That is, the statistic is a **random variable**
- Once we observe the sample, we know the value of the characteristic Y for the sampled units and, in turn, we may compute the value that the statistic takes based on the observed sample

$$\mathbf{y} = (y_1, \dots, y_n)' \rightarrow f(\mathbf{y}) = f(y_1, \dots, y_n)$$

The sample mean

- A parameter of primary interest is frequently represented by the **population mean**
- As stated before, the sample counterpart of the population mean is the **sample mean**, $\bar{Y}$

$$\mu = E(Y) \rightarrow \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

- Clearly, we don't know the value the sample mean will take before we observe the sample
- Further, if a different sample was observed, the specific value of the sample mean would have likely been different
- That is, the sample mean is a random variable and, as such, it can be probabilistically described via a probability distribution function

An illustrative example

- Let's consider a **population** made by 4 units only $\{A, B, C, D\}$
- The characteristic of interest is $Y = \#$ of books read by one unit over a year

Individual	Books read
A	0
B	1
C	3
D	1

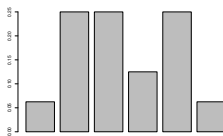
- If we could observe the entire population, we would easily calculate the **mean number of books** read by a unit per year ($\mu = \frac{5}{4} = 1.25$)
- Since we don't usually observe the entire population, the population mean is unknown and represents the parameter of interest

- Suppose we randomly draw a sample of **size 2** from the population
- In our example, the total number of samples comprised of 2 units (with repetition) is $4^2 = 16$
- Each sample has a probability to be selected equal to $1/16$
- Once we draw the sample, we will observe only one of the 16 possible samples

Sample	Individuals	$\bar{x}$
1	AA	0
2	AB	0.5
3	AC	1.5
4	AD	0.5
5	BA	0.5
6	BB	1
7	BC	2
8	BD	1
9	CA	1.5
10	CB	2
11	CC	3
12	CD	2
13	DA	0.5
14	DB	1
15	DC	2
16	DD	1

$\bar{x}$	$p(\bar{x})$
0	$1/16=0.0625$
0.5	$4/16=0.2500$
1	$4/16=0.2500$
1.5	$2/16=0.1250$
2	$4/16=0.2500$
3	$1/16=0.0625$

- Sampling n units from the population and computing the sample mean on the observed units is a random experiment
- Before observing the sample, the value of the sample mean is unknown
- That is, before observing the sample, the sample mean is a random variable
- As such, it is possible to determine the list of all values it can take and the corresponding probabilities before observing the sample
- In our example, the sample mean is a discrete random variable taking value 0 with probability 0.0625 ($1/16$), value 0.5 with probability 0.25 ($4/16$), value 1 with probability 0.25, etc...



Sampling distribution

A **sampling distribution** of a statistic is the probability distribution that specifies probabilities for the possible values the statistic can take on samples of a given size

Being this a proper probability distribution, a way for summarizing it is computing the **mean** and evaluating its variability around such a center via the **variance**

Mean and variance of the sample mean

A characteristic Y has a given mean μ and variance σ^2 in the population
We can evaluate the **mean** of the distribution of the sampling mean

$$\begin{aligned} E[\bar{Y}] &= E\left[\frac{Y_1 + Y_2 + \cdots + Y_n}{n}\right] \\ &= \frac{E[Y_1] + E[Y_2] + \cdots + E[Y_n]}{n} = \frac{n\mu}{n} = \mu \end{aligned}$$

as well as the **variance** (of the sampling mean)

$$\begin{aligned} \text{Var}[\bar{Y}] &= \text{Var}\left[\frac{Y_1 + \cdots + Y_n}{n}\right] && \text{due to independence} \\ &= \frac{\text{Var}[Y_1] + \cdots + \text{Var}[Y_n]}{n^2} \\ &= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \end{aligned}$$

Standard Error of the sample mean

- The spread of a sample statistic around its mean is called **standard error**
- This simply corresponds to the standard deviation of the statistic
- For the sample mean, the standard error is

$$se(\bar{Y}) = \sigma_{\bar{Y}} = \sqrt{Var[\bar{Y}]} = \frac{\sigma}{\sqrt{n}}$$

Interpretation

- The distribution of the sample mean is “centered” on the true μ
- Its standard error becomes smaller for larger sample sizes (i.e., if n increases) and it is always proportional to the population standard deviation σ
- The population standard deviation is a fixed (unknown) quantity; by varying the sample size n , we are able to increase/reduce the standard error of the sample mean
- **Conclusion:** μ is unknown but we know that, increasing n , the sample mean will (overall) get closer to the true value μ

- The result that $\bar{Y}$ converges to μ as n grows indefinitely is called the **law of large numbers**

Law of large numbers

With n independent observations from a distribution with mean μ , for any $\epsilon > 0$, we have that

$$\lim_{n \rightarrow \infty} P(|\bar{Y} - \mu| > \epsilon) = 0$$

Equivalently

$$\lim_{n \rightarrow \infty} P(|\bar{Y} - \mu| < \epsilon) = 1$$

- This type of convergence involves a sequence of random variables instead of fixed values as in ordinary sequences
- To reflect the randomness, the convergence is called **convergence in probability** and is denoted by $\xrightarrow{P}$

Prove all this via simulations

- Assume we are interested in a characteristic Y of the population
- The population mean for such a characteristic is

$$\mu = 50.0$$

- Assume we randomly draw a sample of increasing size

```
> mean(runif(10, 0, 100)); mean(runif(1000, 0, 100)); mean(runif(10000000, 0, 100))  
[1] 23.58847      # sample mean for random sample of n = 10 from uniform [0, 100]  
[1] 51.86637      # sample mean for n = 1000  
[1] 49.99736      # sample mean for n = 10000000 (population mean = 50.0)
```

- As evident, the sample mean gets closer and closer to the population mean μ

Special cases

- The distribution of the sample mean gets narrower (around the population mean) as the sample size increases, but **what is its shape?**
- Since $\bar{Y} = \frac{1}{n} \sum_i Y_i$, it is clear that the shape of its distribution is dictated by the random variable $\sum_i Y_i$
- For **independent observations** from some families of probability distributions, the sampling distribution of $\sum Y_i$ falls in the same family
- Dividing $\sum Y_i$ by n **rescales** the sampling distribution of $\sum Y_i$ to yield the sampling distribution of $\bar{Y}$

- If $Y_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$, we have that $\sum_i Y_i \sim N(n\mu, n\sigma^2)$, so that

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- If $Y_i \stackrel{iid}{\sim} Be(p)$, we have that $\sum_i Y_i \sim Bin(n, p)$, so that

$$\bar{Y} \sim \frac{Bin(n, p)}{n}$$

- If $Y_i \stackrel{iid}{\sim} Pois(\lambda)$ we have that $\sum_i Y_i \sim Pois(n\lambda)$, so that

$$\bar{Y} \sim \frac{Pois(n\lambda)}{n}$$

That is, the sample mean has the same shape of the characteristic of interest in the population, but less spread

Central Limit Theorem, CLT

For a sample size n sufficiently large, we can always determine the **approximate** distribution of $\sum_i Y_i$

Let $Y_1, \dots, Y_n$, be i.i.d. random variables with means all equal to μ and variances all equal to σ^2 . If n is large enough, then the random variable

$$\sum_{i=1}^n Y_i = Y_1 + Y_2 + \dots + Y_n$$

is **approximately** distributed as a **Gaussian** random variable with mean $n\mu$ and variance $n\sigma^2$. That is,

$$\sum_{i=1}^n Y_i \overset{a}{\sim} N(n\mu, n\sigma^2)$$

The convergence of a sequence of cdf's to a limiting cdf is called **convergence in distribution** and is denoted by $\xrightarrow{d}$

- The CLT has a popular standardized version:

$$Z = \frac{\sum_i Y_i - n\mu}{\sigma\sqrt{n}} \stackrel{a}{\sim} N(0, 1)$$

so that

$$p\left(Z < z\right) \approx \Phi(z)$$

- The CLT applies not matter what the specific distribution of the random variables $Y_1, \dots, Y_n$, is
- The only required conditions are that they are iid and $n \rightarrow \infty$

An example

- An elevator can take up to 5 people
- It may break down if the total weight (Y) is more than 300 kg
- What is the probability that the elevator breaks down if 5 people enter and each person comes from a population with mean weight of $\mu = 55$ and variance $\sigma^2 = 9$ (i.e., $Y_i \stackrel{iid}{\sim} N(55, 9)$)?
- From CLT:

$$\sum_{i=1}^5 Y_i \approx N(n\mu, n\sigma^2) = N(275, 45)$$

- Then, we can evaluate

$$P\left(\sum_{i=1}^5 Y_i > 300\right) = P\left(Z > \frac{300 - 275}{\sqrt{45}} = 3.72\right) \approx 0$$

Approximate distribution of the sample mean

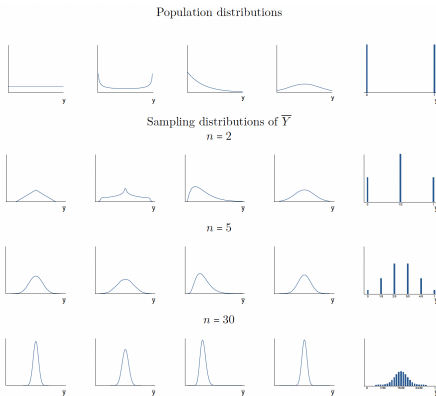
A direct consequence of the CLT is that, if $Y_1, \dots, Y_n$, are iid random variables all having mean and variance equal to μ and σ^2 , respectively, the **sample mean is approximately Gaussian with mean μ and variance σ^2/n** , if n is sufficiently large. That is,

$$\bar{Y} = \frac{1}{n} \sum_i Y_i \stackrel{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

Similarly

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \stackrel{a}{\sim} N(0, 1)$$

As before, the approximate normality of the sample mean applies no matter what the shape of the population distribution is



Remarks

- How large n must be before the sampling distribution is approximately normal depends on the **skewness** of the population distribution
- If the population distribution is **bell-shaped**, then the sampling distribution is bell-shaped for all n
- Unless the skewness is very severe, a sample size of about 30 is sufficient
- Knowing that the sampling distribution of $\bar{Y}$ is approximately normal helps us to find probabilities for possible $\bar{Y}$ values
- For instance, $\bar{Y}$ almost certainly falls within 3 standard deviations ($\sigma_{\bar{Y}}$) from the population mean μ

The sample proportion

- In many applications, the characteristic of interest, Y , is qualitative with only two possible outcomes (yes/no, true/false, etc..)
- That is, we are in the presence of a binary characteristic for which a Bernoulli distribution is a natural choice
- A parameter of primary interest is the proportion of units having the characteristic in the population (the **population proportion** $\rightarrow p$)
- The sample counterpart of the the population proportion is the **sample proportion**

$$\hat{P} = \frac{1}{n} \sum_{i=1}^n Y_i$$

where $Y_1, \dots, Y_n$, denote the iid r.v.s describing the characteristics of sampled units

- As it is clear, the sample proportion is a sample mean for binary variables and, as such, it shares all properties we have till now

Mean, variance, and standard error of the sample proportion

The **mean**, the **variance**, and the **standard error** of $\hat{P}$ are

$$\begin{aligned}\diamond \quad E[\hat{P}] &= E\left[\frac{Y_1 + \cdots + Y_n}{n}\right] \\ &= \frac{E[Y_1] + \cdots + E[Y_n]}{n} = \frac{np}{n} = p,\end{aligned}$$

$$\begin{aligned}\diamond \quad \text{Var}[\hat{P}] &= \text{Var}\left[\frac{Y_1 + \cdots + Y_n}{n}\right] \\ &= \left[\frac{\text{Var}[Y_1] + \cdots + \text{Var}[Y_n]}{n^2}\right] \quad \text{due to independence} \\ &= \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n},\end{aligned}$$

$$\diamond \quad \text{se}(\hat{P}) = \sigma_{\hat{P}} = \sqrt{\text{Var}[\hat{P}]} = \sqrt{\frac{p(1-p)}{n}}$$

Gaussian approximation of Binomial r.v.s

- Consider $Y_i = \{0, 1\}$, so that $Y_i \stackrel{iid}{\sim} Be(p)$, and let $Y = \sum_{i=1}^n Y_i$
- $E(Y) = np$ and $Var(Y) = np(1 - p)$
- As stated before, Y follows a Binomial distribution with parameters n and p (**exact distribution**)

$$Y \sim Bin(n, p)$$

- However, for n sufficiently large, due to the CLT, Y is **approximately normal**, with mean np and variance $np(1 - p)$. That is,

$$Z = \frac{Y - np}{\sqrt{np(1 - p)}} \stackrel{a}{\sim} N(0, 1)$$

Example

- Assume $Y_i = 1$ if voter i supports Biden and $Y_i = 0$ if he/she support Trump
- Of the 17 million adult residents of California who voted, 63.5% voted for Biden
- That is, the proportion of voters supporting Biden in the population (i.e., the population proportion) is $p = 0.635$
- For the exit poll of size $n = 2271$, 65.5% said they voted for Biden
- That is, the proportion of voters supporting Biden in the sample (i.e., the sample proportion) is $\hat{p} = 0.655$
- Had we observed a different sample of size $n = 2271$, almost surely the sample proportion would have been different
- The distribution of $\hat{P}$ is approximately normal with mean equal to 0.625 and standard error equal to $\sqrt{0.625(1 - 0.635)/n}$

